

Isaac Mubiru

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PROFESSIONAL SUMMARY

Backend engineer with 4+ years building distributed systems, real-time data pipelines, and cloud infrastructure at scale. Led zero-downtime migration of 20+ production applications from Parse/MongoDB to Firebase/GCP, designing high-volume API architectures with intelligent failover mechanisms and rate limiting. Experienced in AI/ML integrations including OpenAI API and Google Vision for production systems. Strong track record delivering TypeScript/Node.js backend systems with emphasis on reliability, clean code, and cross-team collaboration in remote-first environments.

CORE COMPETENCIES

TypeScript/Node.js Backend

Distributed Systems

API Gateway Design

Cloud Infrastructure (GCP/AWS)

AI/ML Integrations

Real-time Data Pipelines

Rate Limiting & Failover

Microservices Architecture

WORK EXPERIENCE

MTailor

September 2022 – Present

Software Engineer — US Remote

- **Led end-to-end migration** of 20+ production applications from Parse/MongoDB to Firebase/GCP with zero downtime, reporting directly to the CTO
- **Designed real-time sync architecture** using Node.js and Google Pub/Sub, handling high-volume concurrent requests with intelligent failover mechanisms
- **Built 3D visualization feature** using video overlay and ffmpeg library, increasing buyer conversion rates
- **Saved \$5,000/month** by architecting and executing complete AWS to GCP infrastructure migration
- Implemented Express Shipping feature generating additional \$40 revenue per order

MTailor

March 2022 – July 2022

Contractor — US Remote

- Built migration scripts with complex processing logic for MongoDB to Firestore data transfer
- Authored SDK documentation for engineering team on Firebase integration patterns
- Trained operations team on new Firebase Dashboard and backup procedures

Mind2matter.co

January 2022 – July 2022

Software Engineer — US Remote

- Built backend systems for DeFi applications using Web3 and Node.js
- Delivered React UIs from Figma designs under tight client deadlines

Dr Wealth

August 2021 – January 2022

Software Engineer — Singapore Remote

- **Optimized high-volume data pipelines** querying 2M+ Firestore records for real-time stock price updates
- Extended backend APIs using Firebase Cloud Functions and Express.js with clean, testable code
- Built responsive PWA interfaces using React and Tailwind CSS

CodeBits

January 2020 – July 2021

Software Engineer — Uganda Remote

- **Led team of 4 developers** building distributed legal tech systems for NGOs in Uganda
- **Architected microservices backend** using Apache Kafka, Docker, and Kubernetes with rate limiting and failover patterns
- Built cross-platform mobile app with React Native and Firebase push notifications (iOS & Android)
- Developed USSD service communicating with Node.js backend via gRPC

PROJECTS

Real-time MongoDB ↔ Firestore Sync Pipeline Production

Built live two-way sync supporting parallel production traffic during zero-downtime migration using Google Pub/Sub, retry logic, and idempotent message handling.

Node.js, Google Pub/Sub, MongoDB, Firestore, TypeScript

Healthcare WhatsApp Chatbot with AI Integration AI/ML

Integrated OpenAI API and Google Vision for prescription OCR via WhatsApp Business API, automating healthcare information workflows.

Node.js, OpenAI API, Google Vision API, WhatsApp Business API

Marketplace Recommendation Engine Production

Built collaborative and content-based filtering engine in MongoDB for service provider matching in marketplace application.

Node.js, MongoDB, Python

EDUCATION

Bachelor of Science in Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda

SKILLS

Languages: TypeScript, JavaScript, Python, Go (learning) **Backend:** Node.js, Express.js, Firebase Cloud Functions, gRPC

Cloud: GCP (Pub/Sub, Cloud Functions, Firestore), AWS (S3, EC2, EBS) **Databases:** MongoDB, Firestore, SQL

Infrastructure: Docker, Kubernetes, Apache Kafka **AI/ML:** OpenAI API, Google Vision API **Frontend:** React, React Native, Tailwind CSS